

Cyber Threat Intelligence Pipeline (CCP)

Complex Coursework Project (CCP)
Software Construction and Development
Program: Software Engineering | Section: Blue
PAF-IAST

1. Project Overview

This project is a distributed, event-driven microservices architecture designed to ingest, process, and visualize real-time cyber threat indicators (IoCs). It fetches live malicious IP data, processes it through a message broker, stores it in a relational database, and serves it to a live monitoring dashboard.

2. System Architecture

The system is fully decoupled and consists of four main components:

- **Ingestion Service (Port 8080)**: A Spring Boot producer that connects to the AbuseIPDB REST API, fetches high-risk malicious IP addresses, and publishes the raw JSON payload to an Apache Kafka topic (raw-iocs).
- **Extraction Service (Port 8082)**: A Spring Boot consumer that constantly listens to the Kafka topic. It intercepts the raw JSON, extracts the relevant threat data (IP, Severity Score), and persists it into a MySQL database.
- **Threat API Service (Port 8083)**: A Spring Boot REST API that queries the MySQL database and exposes the processed threat intelligence data to external clients via cross-origin endpoints.
- **SOC Web Dashboard (index.html)**: A frontend UI built with HTML/CSS/Vanilla JS that consumes the Threat API and displays a real-time, auto-updating grid of critical threats.

3. Prerequisites

To run this project locally, you must have the following installed:

- **Java Development Kit (JDK)**: Version 17 or higher

- **Apache Kafka:** Version 4.x (*Configured to run on localhost:9092*)
- **WAMP Server:** For MySQL Database (*Running on port 3306*)
- **IDE:** IntelliJ IDEA or VS Code
- **Maven:** For dependency management

4. Setup & Execution Guide

Step 1: Database Setup

1. Start your WAMP Server.
2. Open phpMyAdmin.
3. Create a new database named threat_intel_db.

(Note: The Extraction Service uses Hibernate ddl-auto=update, so the iocs table will be generated automatically).

Step 2: Kafka Setup

1. Open PowerShell or your terminal.
2. Navigate to your Kafka installation directory.
3. Start the Kafka server using the following command:

```
. \bin\windows\kafka-server-start.bat . \config\server.properties
```

Step 3: Master Startup Sequence

Because this is a microservices architecture, the services must be started in a specific order:

1. **Verify Foundation:** Ensure WAMP (MySQL) and Kafka are actively running.
2. **Start the Database Consumer:** Run ExtractionServiceApplication.java (Port 8082).
3. **Start the API Provider:** Run ThreatApiServiceApplication.java (Port 8083).
4. **Start the Dashboard:** Double-click index.html to open it in your web browser.

5. **Start the Data Producer:** Run **IngestionServiceApplication.java** (Port 8080).

Step 4: Trigger the Pipeline

1. Open a new tab in your web browser.
2. Navigate to the ingestion trigger endpoint: **<http://localhost:8080/ingest>**
3. You should see a **"Success"** message.
4. Return to your SOC Web Dashboard tab and refresh the page.

The grid will instantly populate.